LEARNIX

An explainable, steerable course recommender for degree-program advising

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B.TECH. PROJECT

THE PROBLEM

Course registration leaves students to manually cross-reference transcripts against prerequisites, guess whether a schedule fits, and judge course relevance to their career with no explainable reasoning and no way to adjust what matters to them.

- No ranking by actual career relevance
- No visibility into prerequisite / schedule conflicts before it's too late
- Zero explanation — recommendations, where they exist, are a black box

OUR APPROACH

A **content-based, multi-criteria** recommender. Every course and every career target carries the same 5-axis skill vector (ML · Statistics · Systems · Algorithms · Practice). Each eligible course is scored live against the student's actual transcript-derived skills, declared goal, interests, schedule and cost — then explained in plain language.

Transcript → Skill Vector → 5 Subscores → Weighted Match % → Plain-Language Reason

TECH STACK

Next.js 16 · TypeScript · React 19 · Tailwind v4 · PostgreSQL · Prisma · Auth.js v5 · Zod · Zustand · Vitest

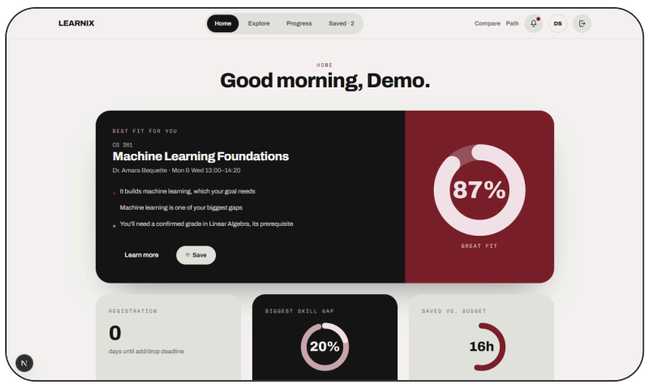
FIVE EXPLAINABLE SUBSCORES

- **Goals** — skill-vector similarity to the career target, boosted by stated interests.
- **Gap** — weighted by how far current skills fall short of the target; closing a bigger gap scores higher.
- **History** — halved if a checkable prerequisite isn't confirmed on the transcript.
- **Schedule** — discounted if the course eats an outsized share of the weekly time budget.
- **Cost** — a fixed credit-value score.

Match % = Σ(subscore × weight) / Σweight — weights are student-adjustable sliders, normalised on every read.

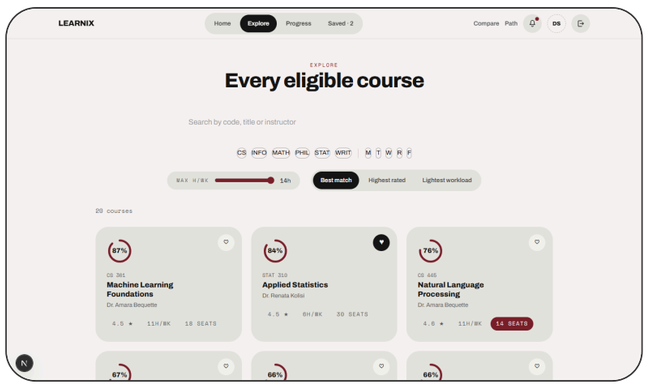
HOME SCREEN

Best-match hero course with live reasoning, plus a ranked rail of alternatives.



EXPLORE SCREEN

Every eligible course, ranked, filterable by department / hours / day.



KEY FEATURES

- **Explore** — filterable, sortable by match / rating / workload.
- **Compare** — two courses side by side across schedule and fit.
- **Progress** — requirement completion and GPA, derived live.
- **Steerable weights** — sliders re-rank everything live.
- **Conflict detection** — saved courses checked for overlap.

BY THE NUMBERS

9	5	27	11	5
STUDENT-FACING SCREENS	EXPLAINABLE SUBSCORES	PASSING UNIT TESTS	RELATIONAL DATA MODELS	SKILL-VECTOR AXES

FUTURE SCOPE

- Greedy multi-term degree planner across future terms
- Instructor/section-level grade-distribution signals
- Collaborative signals layered atop the explainable base

CONCLUSION

Learnix shows a recommender can be rigorous *and* legible — every ranking traces to a named, adjustable signal. No cold-start problem, no black box: a brand-new student gets reasoned recommendations from day one, and can steer the algorithm itself.